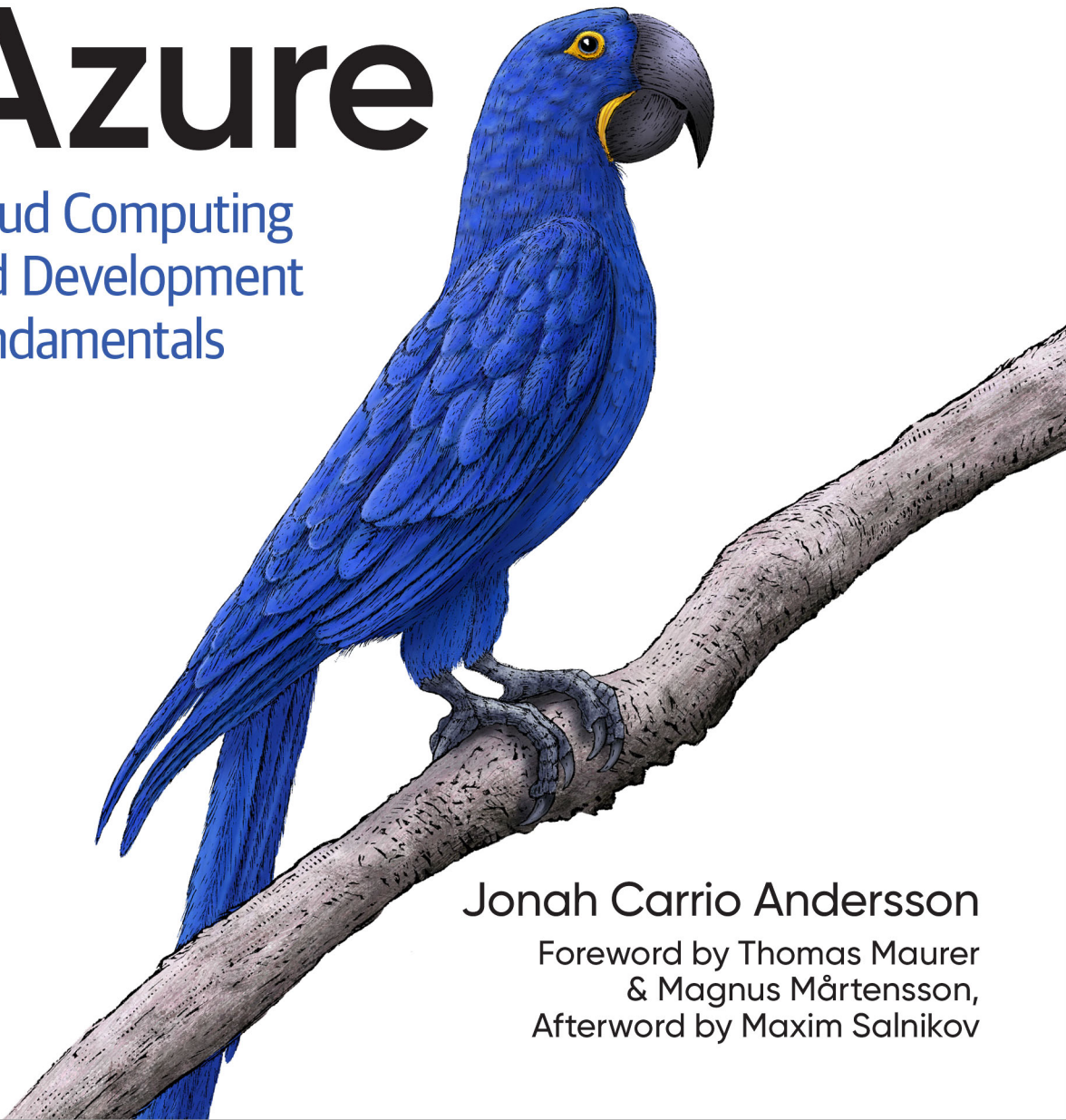


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Learning Microsoft Azure

Cloud Computing
and Development
Fundamentals



Jonah Carrio Andersson

Foreword by Thomas Maurer
& Magnus Mårtensson,
Afterword by Maxim Salnikov

Learning Microsoft Azure

If your organization plans to move from legacy applications to the cloud, implement hybrid or multicloud solutions, or modernize existing applications for scalability, this book is for you. Software developers, solution architects, and cloud engineers will learn fundamental concepts for cloud computing, migration, transformation, and development using Microsoft Azure.

Jonah Carrio Andersson guides you through cloud computing concepts and deployment models, the wide range of modern cloud technologies, application development with Azure, team collaboration services, security services, and cloud migration options in Microsoft Azure.

You'll gain insight into the Microsoft Azure cloud services applicable to different business use cases, software development projects, and modern solutions in the cloud. And you'll dive into Azure container development as well as serverless computing technologies and developer productivity tools that help your IT teams work productively.

- Learn the fundamentals of cloud computing and Microsoft Azure
- Discover which Azure cloud service best suits your particular needs
- Use Azure for different use cases and specific technical requirements
- Start developing cloud services, applications, and solutions using Azure and infrastructure as code
- Learn how to implement governance and cloud security on your IT infrastructure
- Get real-world advice and best practices based on the author's Azure migration experience

"This book is a fantastic journey into Azure, perfect for beginners but also a goldmine for experts. It's the kind of all-around guide that lightens the path for the uninitiated and also serves as a handy reference for seasoned professionals."

—Peter T. Lee
Managing Delivery Architect,
Capgemini America, Inc.

Jonah Carrio Andersson is a Microsoft Most Valuable Professional for Azure and a Microsoft Certified Trainer. She works as a senior Azure consultant focusing on cloud development with .NET and cloud infrastructure within Azure and is a DevOps engineer lead/architect. Based in Sweden, Jonah hosts the podcast *Extend Women in Tech* and is active in the tech community as an international public speaker.

WEB OPS / CLOUD COMPUTING

US \$65.99 CAN \$82.99

ISBN: 978-1-098-11332-2



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Praise for *Learning Microsoft Azure*

Learning Microsoft Azure is a great resource for anyone seeking to advance their knowledge in the vast ecosystem of Azure. This book navigates the entire spectrum of Azure services, catering equally to novices and veterans in the field.

Jonah's meticulous approach in detailing Azure's components mixes architectural insights with practical examples, deepening the reader's understanding. The inclusion of skill assessments ensures that learners are not only consuming information but actively engaging with it.

What sets this book apart is its coherent explanation of how the many Azure services connect, demystifying complex terminology and providing guidance on service utilization. It's clear that Jonah has a passion for making others grow, anyone who has met Jonah will feel that passion while reading the book.

I recommend *Learning Microsoft Azure* to anyone aspiring to broaden their Azure skills.

It's not just a book; it's a comprehensive guide that prepares you for the future of cloud computing.

—Mathias Olausson, CTO at Solidify, Microsoft MVP

This book is a fantastic journey into Azure, perfect for beginners but also a goldmine for experts. It's the kind of all-around guide that lightens the path for the uninitiated and also serves as a handy reference for seasoned professionals.

—Peter T. Lee, Managing Delivery Architect,
Capgemini America, Inc.

Azure is such a vast platform that it may seem only an expert could comprehend its many components. In *Learning Microsoft Azure*, Jonah Andersson masterfully relates the entire stack for a general technical audience. The book is carefully constructed yet easy to read, delivering crucial knowledge about Azure and cloud computing more generally throughout. Data professionals like me will particularly benefit from understanding Azure's machine learning and AI capabilities. I recommend *Learning Microsoft Azure* to any tech professional seeking to gain a hands-on, technical understanding of cloud computing through Azure.

—George Mount, *Data analyst and educator
at Stringfest Analytics*

This is the book I wish I had when I started my journey with Microsoft Azure. Jonah has a fantastic ability to simplify complex technical concepts, making them easy to understand.

Regardless of whether you are a beginner or a seasoned Cloud Engineer, this book will help you understand and learn Microsoft Azure to the fullest, enabling you to benefit from its full range of features and capabilities.

—Nikolaos Delis, *Senior Cloud Engineer and Cloud Evangelist,
Helo AB*

In *Learning Microsoft Azure*, Jonah delivers a comprehensive and accessible book for those working in cloud computing. With clear explanations, this book is a great starting point for beginners and also serves as a handy reference for experienced professionals.

It is an indispensable asset that presents insightful use cases and breaks down complex concepts so they can be easily understood regardless of how much experience the reader has. Whether you're just getting started or aiming for advanced knowledge, this book is a valuable resource for understanding the basics of Azure.

—Priyanshu Jimish, *Technical Evangelist, Virta Technologies*

Jonah Andersson's *Learning Microsoft Azure* thoroughly covers the core facets of Azure but also delves deeply into the nuances of integration, DevOps, and governance. An indispensable resource for any engineer aiming for comprehensive mastery over Azure.

—Dr. Milan Milanović, *CTO and Microsoft MVP*

Writing a review for a book is never an easy task, especially when the book is as comprehensive and useful as this book. *Learning Microsoft Azure* contains practical information on Azure Cloud services, Azure DevOps pipelines, Azure networking, containerization, AI and more.

This book demonstrates Jonah Andersson's superior training skills and knowledge on cloud computing. It provides you with broad, complete and practical information to implement your solutions in Microsoft Azure.

—Reza Salehi, Consultant, Author of
Azure Cookbook

It doesn't matter if you are already experienced or just starting to learn Azure and want to gain a deeper understanding. This book pulls everything you need together and is thoughtfully organized, making it easy to learn.

—Lior Yantovski, DevOps Tech Lead, AT&T

This book is an incredible resource for those who want to learn Microsoft Azure and cloud concepts, spanning from beginner to advanced level. Reading this book will help you implement cloud solutions in real time projects. Jonah Andersson endeavored to cover each and every topic with detailed information and an accessible way of writing.

—Sagar Rastogi, Technical Architect, Tata Consultancy Services

Learning Microsoft Azure by Jonah Carrio Andersson is an exceptional guide that offers a comprehensive and insightful journey through the intricate world of Microsoft Azure. Andersson's expertise shines through, providing readers with a well-structured and detailed exploration of cloud computing, Azure services, AI, IoT, and DevSecOps, making it an indispensable resource for anyone looking to master Azure technologies.

Her clear explanations and practical insights make this book a valuable asset for both beginners and experienced professionals seeking to harness the power of Azure for their projects and career development.

—Amit Dass, Data Architecture Manager, MCT
Certified, Microsoft Azure Data Architect and Lead Data Engineer,
Certified Professional Data Engineer and Architect in GCP, Azure, and AWS (Multi-cloud)

A very well-written, well-researched and engaging book on Azure.
The knowledge and passion of the author really shines through!

Learning Microsoft Azure works well for people with technical and non-technical backgrounds alike. It will also help anyone who wants to achieve Azure certifications. Highly recommended to anyone looking to develop a solid foundation in Azure.

—Andrew Urwin, Microsoft Azure MVP and
Director of Platform and DX at Clue Software

Jonah has not only captured and eloquently explained the concepts pertaining to cloud and Azure fundamentals, but she has brought together the building blocks needed for developers to create apps and deploy services on Azure. She has captured what is necessary to get a start on DevOps on Azure that can translate to other cloud providers as well. Her passion for teaching others shines in this text!

—Dwayne Natwick, ISC2, Microsoft, and CompTIA trainer and
consultant; CEO, Captain Hyper-scaler, LLC

Jonah Andersson's book isn't just a guide; it's a treasure map to unlock the full potential of Microsoft Azure. With its comprehensive insights into cloud migration, development, and deployment, it's a must-read for adventurers in the Azure realm.

—Saeid Dahl, Azure Tech Leader, Sr. Solution Architect
at WeSafe, and Microsoft Learning Expert

The Azure platform can be daunting for beginners, however, in this book, Jonah has managed to make learning the fundamentals easy and understandable. By enriching the technical details with many practical and personal experiences she has brought the examples to life and helped us see how they can be used in the real world. The breadth of her knowledge makes this book accessible to both people just starting out and to those who are more experienced and looking to expand their knowledge.

—John Kilmister, Software Architect and Microsoft Azure MVP

Jonah Andersson's book is an essential read for any IT professional looking to harness the full potential of cloud computing. With its comprehensive coverage, practical examples, and expert insights, this book elevates the reader's understanding of Azure to new heights, covering topics like cloud-native development, DevOps, migration strategies, and much more!

—Freek Berson, Principal Product Manager at Parallels (Alludo)
and Microsoft MVP

In *Learning Microsoft Azure*, Jonah Andersson brilliantly demystifies Microsoft's Azure platform, unveiling its full potential. A must-read guide for unlocking the limitless possibilities of cloud technology.

—Gerald Versluis, Senior Software Engineer at Microsoft

Jonah Andersson's book, *Learning Microsoft Azure*, is a beacon in the vast and ever-expanding landscape of Azure. This book skillfully guides beginners through important cloud concepts, making the immense world of Azure accessible and navigable.

—Michael John Peña, Technical Director at Playtime Solutions
and Microsoft Azure MVP

When learning about the cloud, it can be overwhelming as there are so many services, design patterns, and technical best practices that vary between workloads. Jonah has done an excellent job breaking down the complex facets of Microsoft Azure, from storage to compute and everything in between. I would recommend this book as the “go-to” reference guide for anyone deploying Azure services and building out performant cloud architectures.

—Colby T. Ford, Ph.D., Founder of Tuple—The Cloud Genomics
Company, Microsoft MVP (Azure), and Author of
Genomics in the Azure Cloud

Jonah guides the reader through a clear narrative to learn about Microsoft Azure, from the fundamental concepts of service models and the core concepts of Azure, through to compute, databases, AI services, and governance. This is the book that all those looking to begin their Azure cloud journey, and to get a breadth of knowledge, should be reaching for.

—Jacqui Read, software architect and Author of
Communication Patterns: A Guide for Developers and Architects

Jonah Andersson's book, *Learning Microsoft Azure*, of which I had the honor to be one of the technical reviewers, is an outstanding handbook for anyone who wants to get involved or even just comprehend what cloud computing is, understand specifically what Microsoft Azure is, and navigate what it can offer. Jonah's expertise is apparent in every chapter as she helps the reader learn about each service offered while sharing her experience.

—George Chrysovalantis Grammatikos, Microsoft Azure MVP
and Azure Solutions Architect, Tisski Ltd.

Jonah Andersson's *Learning Microsoft Azure* is a masterful guide that transforms complexity into clarity, presenting the ins and outs of cloud infrastructure with precision and practicality. A must-read for anyone aspiring to excel in the cloud computing sphere.

—Lars Klint, *Principal Advocate, Arkahna and Microsoft Azure MVP*

Learning Microsoft Azure is a comprehensive guide for beginners that want to start embracing the Azure cloud, but also a great guide for cloud experts and solution architects that want to improve their knowledge on this platform. It's a nice read that explains all the different services, from infrastructure to development. I recommend it to everyone that loves the Microsoft cloud.

—Stefano Demiliani, *Microsoft MVP—Azure and Business Applications, CTO at EID NAV-lab Group*

Jonah Andersson is a dynamic community advocacy leader, continually inspiring both current and future community members with her unwavering passion and dedication.

Her book *Learning Microsoft Azure* expertly weaves a comprehensive overview, delivering Azure's fundamentals with precision and depth, making it an invaluable technical companion.

—Kevin Evans, *a.k.a the NETRUNNER, Cloud Solution Architect, Microsoft*

Jonah Andersson has done a fantastic job of distilling complex technical topics into an approachable, easy to read book for technical professionals of all levels. *Learning Microsoft Azure* will provide readers with an opportunity to learn critical cloud computing concepts as well as develop fundamental hands-on skills with Microsoft Azure.

—Dave McCollough, *Chief Technology Officer, Geokey, Inc.*

Learning Microsoft Azure is an exceptionally well-structured and comprehensive guide that provides a fantastic introduction to the diverse functionalities of the Azure platform.

As a freelance developer I see a lot of different tech stacks and that's why I really appreciate good sources of information that I can lean on in my work. This book really hits the spot for me. It is well-written, well organized and the author's reputation for being incredibly supportive and empowering to fellow developers really shines through. Highly recommended for those seeking to unlock their potential in Azure!

—Steffen Jørgensen, *IT Consultant at Sigma Solid*

Learning Microsoft Azure

*Cloud Computing
and Development Fundamentals*

Jonah Carrio Andersson

Beijing • Boston • Farnham • Sebastopol • Tokyo

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Learning Microsoft Azure

by Jonah Carrio Andersson

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Printed in the United States of America.

Published by O'Reilly Media, Inc., 1005 Gravenstein Highway North, Sebastopol, CA 95472.

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Acquisition Editor: Megan Laddusaw

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Proofreader: Charles Roumeliotis

Indexer: nSight, Inc.

Interior Designer: David Futato

Cover Designer: Karen Montgomery

Illustrator: Kate Dullea

December 2023: First Edition

Revision History for the First Edition

2023-11-17: First Release

See <http://oreilly.com/catalog/errata.csp?isbn=9781098113322> for release details.

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978-1-098-11332-2

[LSI]

I would like to dedicate this first book I have ever written to my late and beloved mother, Joan, who died of breast cancer in 2004. Back then at the age of 18, I learned to take my journey in life with resiliency and courage thanks to my mother, who placed her trust and confidence in me. I hope she sees how far I've reached in this life and what I became because of her inspiration and motivation.

I also want to dedicate this book to my family in the Philippines. I devoted most of my life to taking care of my younger siblings and family, and they are always there to morally support me in every exciting journey I take in life and my career.

To my family in Sweden: Thank you to my husband Gösta and to my four-legged best friend Lorentz, who, behind the scenes, provided patience and moral support, and reminded me to take breaks from this book project.

Finally, to all my followers and friends who show support in every adventure and journey I take in life, my career, and even my community engagements.

Thank you so much, Tack så mycket (Swedish), Maraming Salamat (Tagalog)

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Foreword by Magnus Mårtensson

It is not an overstatement to say that *Learning Microsoft Azure* delivers one of the most important current IT skillsets in the market. This might be one of *the* most significant books you, on the way to the cloud, or as a learner already in the cloud, can read right now. Without cloud platforms, modern cloud-based services such as AI and large-scale batch computing, which are crucial for today's businesses, would not exist. There simply are not enough experts in the market that have the broad fundamental cloud platform skills and expertise needed to build these services. The book you hold in your hand is a comprehensive guide that covers the scope of what Azure, fundamentally, is about. These pages provide a solid leg up toward learning true cloud mastery, and the content provides strong competitive information for today's cloud-based IT landscape.

Writing this sort of book, covering so many different but related topics, and succeeding in boiling them down to a consumable size, is no small feat. Honored to write this foreword, I first thought—I could have written this book! As I read, I realized my perspective is not appropriate for authoring a terrific book about learning Azure. I have been an Azure expert and MVP since the first days of Azure, when the name was *Windows Azure*, and I know that it is quite challenging to explain to beginners what Azure is about. You must know what they need to learn and how they see the cloud in order to help them along. The person very much correctly placed to author this work is my friend and superstar Jonah. What she has accomplished with this book is right on the money. She clarifies a wide range of challenging topics in a way that someone starting their cloud journey will be able to consume. Even if you have experience with the cloud or are an expert, you'll still find value in this comprehensive resource due to its breadth and the resources and examples it provides. What Jonah has done is immensely difficult to pull off, *experto crede*, and she has delivered it with technical elegance.

Jonah is a shooting star in the tech space, with years of hard-won technical expertise, and her speaking engagements at conferences feel both strong and natural. As a leader in the Azure community space in Sweden she has turned her natural curiosity

and her work ethic into a set of skills she willingly shares with everyone who is eager to transition to the cloud or expand their migration. It's come as no surprise that Jonah has funneled her energy and passion into a book about Azure.

Logically, the book starts with covering the fundamentals, including what the cloud is and what the Azure cloud is. Through my experience, I have seen that nontechnical business leaders need to understand these foundational concepts in order to map their business needs to the benefits a cloud migration can offer. Beyond business leaders, everyone in a company focusing on cloud should read the first part of the book to lessen the gap between technical and nontechnical backgrounds. Strategically, all companies need to appreciate that the lofty goals of AI, ML, big data, and IoT, which are covered in **Part III**, stand on the foundations of Azure laid out in **Part II** of the book. Accordingly, focusing on your fundamental technical understanding is pivotal to building the future cloud services that will win business in a competitive marketplace. Using the cloud instead of traditional IT will quite likely lead to good results, but there is no guarantee that they will be great. Rather, greatness comes from how you use the cloud and what tools you employ to make your cloud usage truly effective. This includes practices like automation, autoscaling, and effective operations, to name a few. **Part IV** dives into the practices that take you from a good cloud to a great cloud, and **Part V** completes the cloud tour with the tools that are not only in the cloud but are needed to complement the cloud so you can take optimal and full advantage of the power of the cloud.

The name of the game in the cloud is knowledge. The more you know the better your Azure solutions will be! Red Adair said, "If you think it's expensive to hire a professional to do the job, wait until you hire an amateur." That is why this book, which offers a far-reaching overview of the professional cloud landscape, is so useful. It will help you to see a clearer picture and to pierce those cloud veils so that you can find the rainbow in the clouds and the gold at the end of it!

Think of this book as a fundamental handbook to the Azure cloud. You can never arrive at the end of the cloud, because clouds continue to change shape forever. What you can do instead is use this book to gain an understanding of the nature of cloud computing. With this book you will become aware of all of the cloud's distinct characteristics and qualities. Guided by these pages you can take decisive steps straight toward a productive cloud. Congratulations on deciding to focus on learning Microsoft Azure, because you will be ahead of the game after reading this book.

— Magnus Mårtensson
*Loftysoft CEO, Azure MVP since the start of Azure,
Microsoft Regional Director, expert consultant and
advisor, global trainer, and keynote holder
August 2023*

Foreword by Thomas Maurer

Cloud computing is one of the most important and transformative technologies of our time. It enables organizations to innovate faster, scale more efficiently, and reduce costs while delivering better services and experiences to their customers and stakeholders. Microsoft Azure is a leading, open, flexible, enterprise-grade cloud platform that offers more than 200 products and services for building, deploying, and managing cloud applications and solutions. It supports a wide range of programming languages, frameworks, tools, and operating systems. It also integrates with other Microsoft products and services, such as Office 365, Power Platform, GitHub, and Visual Studio. Due to the breadth of its services, Microsoft Azure enables you to create anything from simple websites to complex enterprise applications, from mobile apps to artificial intelligence solutions, from data analytics to Internet of Things scenarios.

In this book, you will learn the fundamentals of cloud computing and Microsoft Azure from an experienced and passionate author who has been working with Azure for many years. Jonah Andersson is not only a software engineer and a DevOps leader but also a Microsoft Most Valuable Professional (MVP) and a community advocate who shares her knowledge and expertise with others through speaking, mentoring, and writing. She has been involved in several cloud migration and transformation projects, and she knows the challenges and opportunities that come with moving to the cloud. I have personally known Jonah for over a decade as a valued member in the Microsoft MVP program, industry leading Azure expert, and friend.

This book is designed to help you get started with Microsoft Azure and gain confidence in using its various services and features. You will learn how to choose the right cloud deployment model for your needs, how to migrate your existing applications or create new ones using Azure services, how to collaborate with your team using Azure DevOps, how to secure your cloud resources using Azure security services, and how to leverage the power of Azure serverless, IoT, and cognitive services to enhance your cloud solutions.

Whether you are a beginner or an intermediate user of Microsoft Azure, you will find this book useful and informative. It covers the essential topics and concepts that you need to know to succeed in your cloud journey. It also provides practical examples and exercises that you can follow along with to reinforce your learning. By reading this book, you will gain a solid foundation of cloud computing and Microsoft Azure that will help you advance your career and achieve your goals.

I highly recommend this book to anyone who wants to learn Microsoft Azure and cloud computing from a trusted and experienced author. Jonah Andersson has done a great job of writing a clear, concise, and engaging book that will guide you through the fundamentals of Microsoft Azure. I hope you enjoy reading this book as much as I did.

— *Thomas Maurer,*
Senior Program Manager and Chief Evangelist,
Azure Hybrid Microsoft
August 2023

Preface

Greetings from the Author

Hello, hej, and mabuhay to you, the reader of *Learning Microsoft Azure*!

As the author, I want to thank you for picking up this book and choosing to read it.

I appreciate that you invested time, money, and effort to read this book to help you learn about Microsoft Azure. Our time is precious and investing time in learning is golden. I hope you find this book a valuable resource for your learning journey. Application development, DevOps, and engineering with cloud technologies have been the most exciting experiences in this modern era of artificial intelligence. We, as innovators and builders of these technologies play a great role of the future of our modernization and digitalization.

We have gone from the age of huge steel mainframe computers to massive virtual modern resources and technologies on the cloud. I remember when I was studying computer science, I had to use several of those low-memory 3½-inch floppy disks to save a programming project on the computer. Today, we don't have to worry about that. As much as the industrial revolution forever changed the manufacturing industry and our access to consumer goods, digitization through cloud computing has transformed how we live and work.

Microsoft Azure is a cloud computing platform close to my heart. For many years, I have been developing applications hosted on premises and on the cloud, working with the modern DevOps practices we have today. Working several years in the IT industry in different roles, I have learned a lot and gained significant hands-on experience, both technical and soft skills, and I am inspired to share knowledge about Azure.

Azure is a global cloud provider serving millions of organizations, customers, and users who are building modern applications. This cloud platform brings powerful

benefits, not just to the business, but to diverse members of the organization, from leadership to project managers, clients, and engineering teams.

Developing and maintaining enterprise systems is my daily work routine. The experience I gained in the software engineering and working in the IT consulting industry in different areas has helped broaden my technical knowledge and toolbox, which I am now able to share with you.

The Cloud Migration Journey to Azure That Leads Me to You

I was once involved in a cloud migration project to Azure to help develop, migrate and rehost an old legacy .NET application hosted on on-premises servers. That legacy application was becoming outdated; it was also not documented and had different code patterns and styles developed by different types of developers. Alongside a small team assisting me, I had full responsibility for moving it to a cloud computing platform. Although it is not really Agile to do it alone without a dedicated team, it was part of my job, and I also wanted to take the opportunity of learning from new challenges. So, I could say that I was like a superwoman on a cloud migration mission.

From designing the system architecture to restructuring the databases, fixing the technical debts, fixing data quality issues, programming, and even creating documentation, I put my heart into it. But that cloud migration project was a fiasco.

After trying different migration alternatives (lift and shift, refactoring code, and re-architecting the infrastructure), we determined that the system needed to be rebuilt as though new. Unfortunately, that project was stopped just when we were close to success. It was not deployed to production because the organization lacked a cloud migration strategy, awareness of the significant benefits of using cloud technologies, and knowledge about the cloud, among many other organizational factors.

It was a project that didn't make it to the finish line. Still, it was one of the most significant experiences in my cloud engineering career. I learned a lot from that migration to cloud project, including the mistakes we made in that project and other lessons to take forward.

Just like any other project, and in life, there are no regrets, just lessons learned.

It is my hope that you as the readers of this book will not commit the same mistakes we did in that cloud migration project. This is one of the reasons why this book exists. It is my hope that learning Microsoft Azure will help you prepare and take smart steps in any cloud migration project you will be involved in.

My cloud migration journey as a developer, and partly the role of being a cloud architect, is one of the reasons I am passionate about sharing knowledge of cloud computing and Azure, especially to those still starting their journey. I inspire knowledge through public speaking at tech conferences, meetups, etc. I also apply what I teach in the career roles I do every day.

It is an honor to spend part of my late nights, after-work hours, and weekends writing this book for you. I hope that the learning you will explore and gain from this book will come in handy in your work, career development, cloud migration journey, and contribution to your organization's cloud engineering projects.

One day the information you learn about Azure might be outdated, but you are on the right track starting a great journey with me through this book.

The entire book has a lot of learning and inspiration in store for you. It is my hope that after you read this book, you will want to learn more by doing and applying it to your existing and future projects!

Why I Wrote This Book

My cloud migration journey as a developer is the primary reason I wrote this book. I also want to inspire and help others to truly plan and get ready to move their workloads and applications to the cloud. Although the clouds are really exciting and beautiful, we cannot just move up there! We need to prepare because by failing to prepare we are creating risks to fail.

When I was studying computer science in the 1990s, my first thesis was about how the invention of internet technology helped local communities and our society. I was curious to finish my thesis and truly understand what the internet was really about and how people adapted to its early phase. The result of my thesis research from community surveys I conducted inspired me to study more about it, which led me to my journey of becoming a tech enthusiast and programmer. Since then, I have become fascinated with how new technologies affect us in our daily routines and work.

As a developer (focused mainly on Microsoft technologies) and cloud platform and DevOps engineer I create, build, and develop technical solutions with modern technologies, including the cloud services offered by Azure. I collaborate in application development, cloud infrastructure management, and automation with DevSecOps.

The more I collaborate with different teams in different areas and the more problems I solve, the more technical skills and knowledge I gain to keep me up-to-date with evolving modern technologies.

I am passionate about sharing this knowledge with others, especially those who want to gain fundamental learning of cloud computing and Microsoft Azure.

This book will help IT professionals, project teams, software developers, and cloud engineers choose the appropriate cloud service to use in Microsoft Azure. This knowledge will help solve your organization's customer use cases and business requirements.

Who Should Read This Book

This book is an essential learning reference book for anybody who wants to learn about the vital cloud concepts and cloud computing services provided by Microsoft Azure, regardless if you are a beginner or at an intermediate level in this field.

The content of this book will also help you gain the technical knowledge to plan, design, and develop applications and modern technological solutions and migrate existing workloads and systems to the cloud using Azure.

This book is for anybody with a technical background with career roles like software developers, cloud engineers, or cloud solution architects. Management or leadership roles in an IT organization, such as IT project managers, technical sales managers, and scrum masters, would also benefit from learning the concepts of cloud computing in Azure. Also, teams working with traditional and on-premises legacy applications or systems will gain essential insight into designing and developing solutions in the cloud platform through the Microsoft Azure service.

This book is also ideal for IT professionals, software developers, and aspiring cloud engineers who want to earn Microsoft Certifications for Azure, such as [AZ-900, Microsoft Azure Fundamentals](#); [AZ-204 Developing Solutions for Microsoft Azure](#), [AZ-400, Designing and Implementing Microsoft DevOps Solutions](#); and [AZ-305, Designing Microsoft Azure Infrastructure Solutions](#). Please note that the exam names of the Microsoft Azure certifications may change at any time.

This book will give you the concepts you need to get started with cloud-native development, administration, data engineering, DevOps, analytics, migration strategies, and other exciting solutions in Azure.

What You Will Learn

By the end of this book, you will understand the following:

- Essential and fundamentals concepts about cloud computing and its benefits to you as a developer and/or IT professional and to your organization
- The core concepts and fundamenals of Microsoft Azure as a public cloud provider to help you build modern cloud solutions and applications
- The different Microsoft Azure technologies that will help you and your organization develop, transform, modernize, and migrate to modern cloud environments

- A comprehensive overview of the different cloud technologies in Microsoft Azure that will help you choose the right cloud service for your demands, use cases, software development, and cloud development projects
- A jump-start guide on how you can start developing cloud solutions and accelerate your career as an Azure developer or DevOps, cloud, or platform engineer
- Start developing modern cloud services, applications, and solutions in Microsoft Azure environments using your desired and supported programming languages, frameworks, and tools
- Learn about the different cloud security tools, including identity and access management (IAM), in Azure to secure your cloud resources, applications, and users
- Learn how you can integrate cloud technologies with other services, APIs, and third-party services
- Gain knowledge about services in Azure that will help you implement cloud security and DevSecOps practices in your organization
- Get practical options and learn from best practices on the important things you need to consider when you migrate existing legacy applications to a cloud platform like Microsoft Azure using different migration tools and services
- Get information about the recent tools and cloud adoption frameworks in Azure that will help you in multi-cloud or hybrid cloud environments
- Develop with Azure by learning the best practices, different programming languages, and modern developer tools like Azure Developer CLI (azd), GitHub CLI, GitHub Copilot, GitHub Codespaces, and Microsoft Dev Box—all of which you can use to build cloud-native applications and solutions such as infrastructure as code (IaC), serverless, containers, and other cloud technologies in Azure.



Learning Microsoft Azure is your guide as you work with Microsoft Azure. I believe in the **idea of learning by doing**.

Check out a blog post I wrote about this topic: “**Continuous Learning — An Integral Part of A Programmer’s Development**”.

Invest time in learning the fundamental concepts and doing some hands-on work.

Navigating This Book

Part I: Cloud Computing and Microsoft Azure Fundamentals

In this book's introductory part, you will learn about cloud computing and Microsoft Azure fundamentals.

Chapter 1, “Cloud Computing Fundamentals”, provides an introduction to the essential concepts of cloud computing, how it works, the different types of deployment models, understanding the types of cloud, what CapEx and OpEx are in cloud computing, and the benefits of utilizing cloud computing in businesses, IT organizations, society, and software engineering.

Chapter 2, “Microsoft Azure Fundamentals”, focuses more on the theoretical and technical concepts of Microsoft Azure as a public cloud platform. Learn about the Microsoft Azure core components and the different cloud services categorized by their purpose.

By the end of **Part I**, you will gain vital knowledge and foundational concepts of cloud computing and Microsoft Azure.

Part II: Compute, Networking, Storage, and Databases

This second part of the book and its chapters focus on the different technologies in Microsoft Azure, grouped into categories.

Chapter 3, “Microsoft Azure Cloud Compute Services”, explores some of the Microsoft Azure compute services such as Azure Virtual Machines; container services like Azure Container Instances, Azure Container Registry, and Azure Container Apps; Azure App Services for web and mobile applications; serverless cloud solutions with Azure Functions; Azure Static Web Apps; and more.

Chapter 4, “Microsoft Azure Cloud Networking”, covers cloud networking and services in Microsoft Azure, including Azure VNet, DNS, Azure Firewall, Azure Front Door, ExpressRoute, Virtual Network, VPN Gateway, Application Gateway, Load Balancer, Internet Analyzer, and more. This chapter briefly mentions Azure Orbital, a fully managed ground station as a service (GSaaS) solution in Azure.

Chapter 5, “Microsoft Azure Cloud Storage and Databases”, provides a technical overview of the different cloud storage options and databases (both SQL and NoSQL) in Microsoft Azure. You will learn about cloud storage concepts and services, create databases, and find applicable best practices for Azure SQL Databases, Azure Cosmos DB, Database for MySQL, Azure SQL Servers, Redis Cache in Azure, Azure Storage, Data Share, and Manage Disks.

By the end of **Part II**, you will be ready to start developing and building applications for the cloud using the compute, data storage, database, and networking options in Azure.

Part III: Artificial Intelligence (AI), Machine Learning (ML), Big Data, IoT, and Security

Chapter 6, “Artificial Intelligence, Machine Learning, and Cognitive Services in Azure”, focuses on the valuable concepts you need to know about artificial intelligence (AI), machine learning, Azure OpenAI Service, and Cognitive Services in Microsoft Azure. This chapter also covers the importance of responsible and ethical AI.

Chapter 7, “Big Data, Reporting, and Analytics Services in Azure”, explores the Big Data, Reporting, and Analytics Services in Microsoft Azure. It will include what you need to know about data analytics, big data, analytics, and reporting services using Power BI, Azure Stream Analytics, Data Lake Analytics, Azure HD Insights, and Azure Analysis Services. This chapter will help you learn the critical concepts for working with extensive complex data using Azure services and big data tools.

Chapter 8, “Cloud IoT and Maps Services”, covers the Azure solutions for IoT (Internet of Things), Maps Services, and Cognitive Services in Azure. You will learn about Azure IoT Hub, IoT Edge, Azure Maps, Azure Spheres, and Remote Rendering services. You will learn how to use some of the Azure IoT kits available for developers such as the Azure Developer IoT Dev Kit to start IoT development with Microsoft Azure.

Chapter 9, “Azure Security, Identity Management, and DevSecOps”, dives into identity and access management (IAM), compliance, cloud security, and DevSecOps in Microsoft Azure, which is important in protecting and securing your applications and cloud workloads. You will learn about Microsoft Entra ID. Azure security services like Azure Key Vault, Azure Sentinel, Azure Firewall, Microsoft Defender for Cloud will be explored, along with other Azure security monitoring features for networking, data, and compute.

By the end of **Part III**, you will have gained technical knowledge and hands-on experience with the different technologies for developing applications with AI, ML, big data, and cloud security in Microsoft Azure. These will help you choose what Azure technology to use for your business requirements and how to develop solutions with them.

Part IV: Integration, Infrastructure, and DevSecOps

This book's fourth part will teach you how to integrate Azure technologies with other services, use infrastructure as code solutions, and set up your application using Azure's deployment technologies.

Chapter 10, “Azure Cloud Integration Services and Tools”, focuses on Microsoft Azure integration services. In this chapter, you will learn about the different cloud technologies you can integrate with your cloud-native applications, including existing ones (on the cloud or on premises). This chapter covers Azure API Management, Azure Logic Apps, Azure Web PubSub, Azure Service Bus, Azure Event Grid, Azure Logic Apps, etc.

Chapter 11, “Cloud Infrastructure, DevOps, and Monitoring in Azure”, guides you on how to develop systems or applications with modern DevOps tools. You will learn what DevOps means, and about Azure DevOps, a comprehensive suite for team collaboration for developers and IT operations. You will also learn about automating development processes using CI/CD and source code version control, what Azure Pipelines, GitHub, and Azure DevOps are, and how Azure Monitor and Application Insights are used for troubleshooting Azure resources. You will also learn about other cloud technology services like Azure DevTestLabs, Azure Bicep, ARM templates, and more. You will also learn about infrastructure as code (IaC), policy as code, and configuration as code and how they help with the automation of deployments.

After reading the chapters in **Part IV**, you will have knowledge and development skills to help you work effectively as a cloud developer using cloud integration and automation options. You and your organization's Agile team will also be able to collaborate and work effectively using the great features of Azure DevOps, DevTest Labs, GitHub, etc.

Part V: Governance, Migration, Architecture, and Development Tools

The fifth section of this book will give you essential knowledge and tools for adopting and migrating successfully to the cloud.

Chapter 12, “Cloud Management and Governance in Azure”, walks you through the essential concepts regarding cloud management and cloud governance in Microsoft Azure. For example, it covers automation in the cloud, Azure Advisor, backup using Azure Backup, Azure Blueprints, Azure Policy, Azure Monitor, and other known solutions for hybrid and multi-cloud.

Chapter 13, “Cloud Migration, Hybrid, and Multi-Cloud Solutions in Azure”, discusses cloud migration, cloud transformation, and architectural concepts in Microsoft Azure. You will learn the vital facts needed for adopting, transforming, or migrating to Azure. Find out the best practices and a helpful list of tools you can

use when moving on premises and handling legacy applications. This chapter also highlights the importance of having Microsoft Azure's Well-Architected Framework.

Chapter 14, “Cloud Development Tools for Azure”, is more focused on cloud development with Azure. This chapter discusses the different language and cloud development tools you can use to get started with developing modern and resilient applications and solutions on Azure. It covers IDE tools such as Visual Studio, Visual Studio Code, Azure Cloud Shell, GitHub Copilot, Codespaces, GitHub CLI, Azure Developer CLI (azd), Microsoft Dev Box, and many more. You will learn the different technologies that will prepare you for your projects and career development.

After reading the chapters in **Part V**, you will have an essential understanding of Microsoft Azure cloud governance and management. These are important in designing and developing cloud solutions. You will also capture a broad understanding of the Microsoft Cloud Adoption Framework (CAF), Well-Architected Framework for Azure (WAF), Azure Migrate, and Microsoft Assessment tools that can assist your cloud migration projects. You will also have learned from lessons I learned from a cloud migration project experience in migrating .NET legacy on-premises applications to Azure. You will deepen your knowledge on cloud development using the development and DevOps tools well integrated with Microsoft technologies and Azure. You will also learn some of the different ways to migrate and develop your existing IT infrastructure and on-premises applications as hybrid or multicloud on Azure.

Check Your Knowledge

At the end of each chapter, I will provide a short *Check Your Knowledge* section with a maximum of five learning check questions related to the topics discussed. These questions will be a useful review challenge for you to confirm what you have learned.

Learn by Doing (Try It!)

As a developer, I love the concept of learning by doing. I believe in theoretical learning enhanced by hands-on practices. Therefore, each chapter in this book also ends with a section called *Learn by Doing (Try It!)* where I provided a recommended list of quickstart guides and how-to links to further your hands-on learning related to the topic discussed in the chapter.

Learning Resources and Further Readings

Learning Microsoft Azure aims to provide the most important foundational knowledge you need to know. Considering that we have different levels of experience and knowledge, you probably want to explore the discussed topic in more depth. Therefore, at the end of each chapter, I will provide a short list of recommended learning and further reading resources related to the topics discussed. Most references are

linked to the most recent Microsoft documentation for the specific technology or resource mentioned in this book.

What This Book Is Not

This book is not an advanced level book for each cloud technology service in Microsoft Azure. Hands-on examples are provided but are not intended for those who already work with Azure at highly advanced levels.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

Constant width

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

Constant width bold

Shows commands or other text that should be typed literally by the user.

Constant width italic

Shows text that should be replaced with user-supplied values or by values determined by context.



This element signifies a tip or suggestion.



This element signifies a general note about the topic discussed.



This element indicates a warning or caution.

Using Code Examples

Supplemental material is available at <https://github.com/learningazurebook>. For more information about the book, updated code examples, and blogs from the author, please visit the book's website: <https://learningmicrosoftazure.com>.

If you have a technical question or a problem using the code examples, please email bookquestions@oreilly.com.

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Acknowledgments

I want to acknowledge and thank those who helped, supported, and contributed to making this book a success from start to finish. This book would not have been possible without my family, friends, and the incredible people in the tech and cloud communities who helped and contributed.

Special thanks to my friends in the community who read the online early release version of this book and left some good feedback: Billy Hollis, Lior Yantovski, Peter Lee, Richard Vaughan, Keith Atherton, and many others. Also thanks to the technical reviewers who helped the quality of my book. I appreciate the time invested by Peter De Tender, Håkan Silfvernagel, Micheal John Peña, Nikos Delis, Stefano Demiliani, Peter Lee, Sagar Rastogi, Vladimir Serykh, Szabó Attila, Ian Santillan, George Grammatikos, Dipal Choksi, Alexander Wachtel, Dave McCollough, George Mount, Gerald Versluis, Demiliani Stefano, Robin Smorenburg, and other reviewers that I may have missed mentioning.

To Andrew Urwin and Freek Berson who contributed their input about working with DevOps and infrastructure as code (IaC) with the Bicep Language in Azure in *[Chapter 11](#)*.

Thanks to experts and friends in the community such as John Kilmister (also a technical reviewer), Alexey Polkovnikov, Ryan O'Connell, Håkan Silfvernagel, Adrienne Braganza Tacke, Goran Vuksic, Andrew Urwin, Ezhilarasi Chezhiyan, George Mount, Tiago Costa, Marilag Svennevig, Sasha Kranjac, Hamida Rebai, and Kristina Devochko who provided their quotes at the beginning of each chapter of this book.

Thanks to Jill Leonard, Jennifer Pollock, Megan Laddusaw, Elizabeth Kelly, Suzanne Huston, Helen Codling, Rita Fernando, Beth Richards, Aleeya Rahman, and the rest of the O'Reilly Media team who supported me and helped me when I had questions during the development of this book.

A *big* thanks to the inspiring forewords of Magnus Mårtensson and Thomas Maurer, and afterword of Maxim Salnikov, who also helped in the technical review of this book.

Finally, I would like to thank Karen Montgomery for designing amazing covers for O'Reilly books, including the amazing Hyacinth Macaw cover of this book. I fell in love at first sight with the colors and character of this amazing, gentle giant parrot of the world. If you are curious about the cover, make sure you read the Colophon section at the end of this book.

Special mention to anybody who helped me through the journey of this book that I probably missed and most of all to those who followed and supported my writing journey. Thanks to all of you!

Cloud Computing and Microsoft Azure Fundamentals

In this introductory part of the book, you will learn about the fundamentals of cloud computing and Microsoft Azure. This part provides an introduction to the important concepts of cloud computing, how it works, and the different types of deployment models. It gives you a general perspective to understand the types of cloud computing. This part also focuses on the theoretical and technical concepts of Microsoft Azure as a public cloud platform of Microsoft.

Cloud Computing Fundamentals

Over the last decade, cloud computing has changed the way we build and deploy software. It is now easier than ever to deploy highly scalable, resilient, and secure solutions to a global audience, in many cases at a fraction of the previous cost.

—John Kilmister, Software Architect and Microsoft Azure MVP

What Is Cloud Computing?

Before delving into learning Microsoft Azure, it's essential first to understand cloud computing. Cloud computing is a significant technological innovation that delivers various services through the internet, including web servers, databases, data storage, virtual machines, applications, network infrastructure, security tools, software, and other IT infrastructure. *Cloud computing* refers to the virtual storage and access of data and information over the internet, with the actual computing processing transpiring in the cloud.

Understanding the foundational principles of cloud computing is essential for designing and developing solutions in the cloud. As a public cloud platform, Azure provides secure, scalable, reliable, cost-effective, and easy-to-manage ways to build on the cloud. By leveraging Azure, organizations and users developing with it can take advantage of various cloud services and tools to innovate and solve complex business problems.

As a modern, game-changing technology, cloud computing enables businesses, organizations, and teams to access a wide range of IT resources on demand through the internet. With Azure, we can build and deploy cloud-based solutions that are secure, scalable, and cost-effective.



John McCarthy's Interesting Cloud Computing Prediction

John McCarthy, an American computer scientist and cognitive scientist known as the **father of artificial intelligence**, gave some interesting insights about cloud computing in his speech at MIT's centennial celebration in 1961. He suggested that computing could be sold like a public utility, just like water or electricity.

Cloud computing makes life easier for us. A practical example is the possibility of saving our photos, videos, and files into cloud storage with virtually unlimited capacity instead of keeping them on a local storage device with limited storage. Another benefit is virtualizing web servers and databases instead of having the physical infrastructure or servers in costly data centers.

Software engineering and modern IT innovation are a few significant advantages of the cloud. Adopting and implementing cloud computing provides a fully managed cloud computing infrastructure and services with the benefits of scalability, autoscaling, availability, and performance at a flexible global scale.

The different cloud computing solutions give us tools and capabilities to handle peak loads based on demands at any time on a global scale. This is complicated, expensive, and time-consuming in an on-premises environment. Azure seamlessly handles scenarios like this through **horizontal or vertical scaling** or a combination of both.

Benefits of the Cloud in Software Engineering and IT

Cloud computing and software engineering are both evolving rapidly. The evolution of computing in the cloud, along with innovations like machine learning (ML), the Internet of Things (IoT), edge computing, quantum computing, and big data, has caused an increased demand for skillsets and people who can work in these technologies and platforms in the cloud.

Computing in the cloud enables software developers, engineers, and even IT professionals to create, build, test, and deploy technical cloud solutions productively, effectively, and securely. Software engineering teams still need to be expanded to work and develop on premises, and they encounter technological gaps. They may experience the risks of missing the advantages of developing cloud computing systems.

Engineering teams, business teams, and organizations may risk missing the great features, benefits, and capabilities that cloud computing provides. Cloud engineering offers better speed of development, testing, maintainability, automation, scalability, and so much more.

With the advancement of modern automation processes like **infrastructure as code (IaC)** becoming available for cloud infrastructures, benefits such as infrastructure

automation are making software development easier for developers and DevOps teams. Through the IaC approach, there will be more consistency and routines for configuring systems with the capability of replicating systems to several environments.¹

In addition to this, solutions and **platforms using low-code/no-code** like **Power Apps, Azure Logic Apps, AI Builder**, etc., on the cloud also help IT professionals with fewer programming skills and less experience to build modern and smart applications quickly on demand. Infrastructure as code technologies and low-code/no-code solutions will be discussed later in this book.

To complement the benefits the cloud has for engineering teams, IT managers can easily manage their projects and collaborate with their teams by working Agile with available modern and remote collaboration tools in the cloud. For example, Azure DevOps is an all-in-one collaboration suite used by operations and development teams to plan, build, test, deploy, and monitor applications. It helps cover the entire application lifecycle, including Agile project planning, source code versioning, continuous integration and continuous delivery (CI/CD), testing plans, artifacts, and integrations. We will dive more into Azure DevOps, cloud development integration tools, and infrastructure automation and management in **Chapter 11**.

Digitalization and modernization come with great benefits; however, they also come with challenges. These include the challenges of preparing, transforming, and adapting to fast-changing and evolving technologies. These barriers can be handled by learning the foundations of cloud computing. Instead of having our resources like databases, applications, servers, or infrastructure in physical data centers or on premises, we have these resources on the cloud or the internet.

¹ Kief Morris, 2020, *Infrastructure as Code*, O'Reilly Media, <https://learning.oreilly.com/library/view/infrastructure-as-code/9781098114664>



In cloud computing, a data center is a physical facility that houses many servers and other computing equipment. Consider a data center a place or physical location meant to provide a centralized place for storing and managing servers for databases and applications that users can access remotely over the internet.

Data centers are the backbone of cloud computing. For example, in Azure, they are the infrastructure necessary to deliver cloud-based services, including storage, computing power, and network connectivity. A data center's physical layout and design are planned to ensure high availability, security, and energy efficiency. They are equipped with redundant power and cooling systems, backup generators, and other measures to ensure continuous operation, even during a power outage, bad weather conditions, or any type of disruption. They also employ strict security measures with biometric authentication of whoever enters the premises. Security systems detect intrusion to protect the resources and assets within the data center premises.

Businesses and organizations consider cloud computing technology a good and strategic option because of its speed, reliability, financial savings, productivity, efficiency, security, performance, and more. By the end of this chapter, you will learn the specific benefits cloud computing has for different categories.

Cloud computing has been a popular option for many because of the growing demand and evolution of technological innovations over the past decades. Organizations want to improve and modernize their systems to innovate with the new trends in technology. Computing in the cloud is expanding and continues to grow; our mission to increase sustainability is also one of the great drivers of cloud innovation.²

Cloud Computing Versus Virtualization

Cloud computing and virtualization both create useful virtual environments. Hosting compute and data resources on the cloud is better than a virtual machine for a web server or databases hosted in an on-premises environment. Hosting in the cloud provides efficiency, flexibility, reliability, global scalability, and security.

The cloud is an environment, while virtualization is a technology that enables us to virtualize hardware to create and simulate several machines or dedicated resources. The cloud is an IT environment that pools and shares scalable resources across a network. Cloud environments are created to activate the extraordinary capabilities of cloud computing, such as running workloads within it.

² Bernard Marr, October 25, 2021, "The 5 Biggest Cloud Computing Trends in 2022," Forbes.com, <https://www.forbes.com/sites/bernardmarr/2021/10/25/the-5-biggest-cloud-computing-trends-in-2022/>